The Hidden Cost of Globalization in the Underwater World: Evidence from California's Kelp Forests

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Abstract

This article examines how a maritime regulation, California's Ocean-Going Vessel Fuel Rule (2009), has affected kelp forests, one of the most important aquatic ecosystems along the California coast. That policy led part of the traffic to move offshore, with potential impact on kelp in the regulated area. Combining satellite measures of kelp canopy and biomass with AIS vessel trajectories, we exploit variation in pre-policy shipping exposure across kelp cells. After the rule, kelp improves disproportionately where maritime traffic was highest. A doubling of pre-policy traffic is associated with about 10–11% more kelp. Event studies show no differential pre-trends, with the treatment effect appearing in 2009, the year of implementation. We apply the same design to the 2011 boundary modification around the Channel Islands, which re-concentrated vessel traffic in specific corridors. Kelp declined most in the cells that had the highest pre-modification shipping exposure. JEL: Q57, R41, F18

Keywords: Kelp forests, maritime regulation, emission control areas, difference-in-differences, ecosystem services, AIS, shipping behavior

1 Introduction

Globalization relies heavily on maritime routes that generate substantial negative externalities and hidden costs. The human health consequences of shipping-related air pollution are now well documented (e.g., Klotz and Berazneva, 2022; Hansen-Lewis and Marcus, 2022), but evidence on impacts on marine ecosystems remains comparatively limited

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at the spatial scale relevant to policy. This paper investigates it by quantifying how a major coastal shipping regulation and the associated reconfiguration of traffic impact the kelp forests in California. Such an analysis contributes to documenting the spatial and environmental footprint of maritime activity.

Kelp forests are large biogenic habitats that support coastal biodiversity and provide multiple ecosystem services (Dayton, 1985). As a foundation species (Dayton, 1985), kelp creates a habitat structure that functions as nursery grounds and supports complex food webs in the world, such as in Europe (Christie et al., 2009) and California (Steneck et al., 2002; Carr and Reed, 2019). As a result, the kelp forest has an economic value through fishery production, nutrient cycling, and carbon sequestration, which, according to Eger et al. (2023), can be estimated at between $465 and $562 billion per year worldwide. Finally, because kelp can respond quickly to changing local conditions through rapid growth and recruitment (Krumhansl et al., 2016; Carr and Reed, 2019), these forests provide us with enough variation over a relatively short horizon to study the impact of vessel traffic changes.

We study California's Ocean-Going Vessel Fuel Rule, which established a regulated zone in July 2009, later modified at the end of 2011. The policy was designed to reduce sulfur-related emissions by requiring the use of low-sulfur fuels within the zone.[1] For our setting, the rule altered vessel routing decisions (Klotz and Berazneva, 2022), thereby changing the spatial footprint of maritime activity near kelp forests. In particular, vessel traffic declines after 2009 and re-concentrates after the 2011 boundary change, which provides quasi-experimental variation to study the sensitivity of kelp forests to maritime exposure.

Our contribution is twofold. First, we combine quarterly satellite-based measures of kelp canopy and biomass with AIS-derived shipping exposure. This continuous difference-in-differences design aims to study whether kelp cells that were more exposed to shipping before 2009 benefit the most from this regulation. We find evidence consistent with this prediction. The coefficient on shipping exposure after the implementation of the policy is positive and significant across all specifications.

Second, we apply the same analysis to the 2011 boundary modification around the Channel Islands, which led traffic to re-concentrate in specific corridors. We find that kelp cells more exposed to vessel traffic before 2011 experienced disproportionately worse outcomes after the boundary change. The two results of this paper are thus consistent. Kelp recovers in the areas that benefited most from the 2009 regulation, but deteriorates in the areas most affected by the 2011 re-concentration of traffic.

This analysis contributes to three different literatures. First, it brings marine forests into the economic analysis of deforestation. A growing literature studies how globalization impacts terrestrial forests (e.g., Berman et al., 2023; Abman and Lundberg, 2020; Carreira et al., 2024; Nedoncelle et al., 2024), but underwater forests have received no attention in economics despite providing comparable ecosystem services (habitat provision, carbon sequestration, and nutrient cycling). By analyzing how maritime traffic

[1] Vessels entering the zone must switch from conventional heavy fuel oil, which can contain up to 3.5% sulfur by mass, to marine distillate fuels capped at 0.1%. The switch must be completed before entering the regulated zone (within 24 nautical miles of the California baseline).

affects kelp forests, we extend the deforestation literature that uses causal inference analysis to a new biome and a new mechanism. Second, while experimental and site-specific studies have already been done on kelp (e.g., Johansson et al., 2012), a large spatial analysis (along the California coast) related to a maritime policy has never been conducted. Third, while many economic analyses often focus on the impact of pollution on human health (e.g., Klotz and Berazneva, 2022; Hansen-Lewis and Marcus, 2022), we go here beyond this anthropocentric perspective.

2 Background

The California–Baja California coast hosts extensive kelp forests dominated by canopy-forming species (giant kelp, *Macrocystis pyrifera*, and bull kelp, *Nereocystis luetkeana*). These species form surface (or near-surface) canopies that are detectable with satellite imagery (Bell et al., 2023).

Kelp performance is shaped by ocean temperature, nutrient availability, underwater light (or coastal darkening), as well as localized stressors that can damage fronds, inhibit photosynthesis, or alter community structure (Carr and Reed, 2019).

Maritime activity may affect several of these factors, because vessels release pollutants and disturb the water along their routes. Busy corridors may thus be disproportionately hostile to kelp recruitment and growth. In particular, vessel operations generate chemical and nutrient discharges (greywater and blackwater) that can contribute to eutrophication and oxygen depletion (Lindgren et al., 2016), with potential consequences for coastal vegetation (e.g., Filbee-Dexter and Wernberg, 2018).

A second channel operates through the combustion of heavy fuel oil (HFO), which was used before the implementation of California's emission control area (ECA). HFO combustion generates pyrogenic polycyclic aromatic hydrocarbons (PAHs), which are then found in water and accumulate in coastal sediments (Yunker et al., 2002). Appendix A provides spatial evidence for this channel. We find that PAH concentrations are correlated with our AIS-derived measure of pre-policy vessel traffic intensity (Spearman $\rho = 0.55$; Figures 5 and 6).

Shipping can also introduce contaminants through antifouling practices. Biocidal compounds historically used to reduce hull fouling, including tributyltin (TBT) and copper-based formulations, could enter coastal waters and affect marine organisms (Dafforn et al., 2011; Cima and Varello, 2022). Laboratory evidence confirms that antifouling compounds inhibit kelp photosynthesis and growth (Johansson et al., 2012), and field measurements in Southern California harbors document biocide concentrations that may harm macroalgae (Schiff et al., 2011). Finally, maritime traffic can influence nearshore physical conditions (through wake, turbulence and shading).

As a result, we consider that changes in the spatial distribution and intensity of shipping activity can matter for kelp outcomes. California's Ocean-Going Vessel Fuel Rule, which introduced a regulated zone extending 24 nautical miles off the coast, enables us to test this hypothesis. Under this law, vessels must use low-sulfur marine distillate fuels within the zone. Because compliance increases operating costs within the regulated area, this regulation induces changes in the spatial distribution of traffic. More precisely,

using one-minute AIS data for roughly 20,000 voyages per year off the U.S. west coast, Klotz and Berazneva (2022) document a sharp discontinuity at the July 2009 enforcement date. Container ships reduced fuel consumption within the regulated zone by 33–52%, primarily by rerouting outside its boundary and traveling south of the Channel Islands rather than through the Santa Barbara Channel. Finally, in 2011, the southern boundary was modified around the Channel Islands, affecting where ships concentrate as they approach ports and creating additional variation in maritime exposure.

3 The 2009 Ocean-Going Vessel Fuel Rule

This section describes the data and the continuous difference-in-differences design used to estimate changes in kelp canopy and biomass associated with the 2009 implementation of the regulated zone.

3.1 Data

We combine satellite kelp observations, AIS vessel tracking data, and environmental covariates, all aggregated to a common spatial grid of 10 km hexagonal cells at quarterly frequency.

Kelp outcomes and shipping exposure. Kelp outcomes come from Bell et al. (2023), which provides quarterly cloud-free Landsat-based measures of floating kelp canopy at 30 m resolution along the Pacific coast.[2] We use two variables: (i) kelp canopy area within each pixel (m^2) and (ii) kelp canopy biomass (kg) inferred from validated relationships between surface reflectance and in situ canopy and biomass measurements (Bell et al., 2023). Satellite detection captures surface (or near-surface) canopy and does not observe subsurface kelp.

We overlay a uniform grid of 10 km hexagonal cells covering the study area (the 2009 regulated zone and the 2011 boundary modification area) and aggregate all variables to this common spatial unit at the quarterly frequency.[3]

For each year–quarter, the 30 m pixel-level kelp observations are converted to point geometries using their latitude and longitude coordinates and spatially joined to the hexagonal grid. Within each hexagon i and quarter t, we compute the sum and mean of kelp canopy area (m^2) and biomass (kg), as well as the count of kelp pixels.

[2]The underlying imagery consists of Landsat Collection 1 Level-2 Surface Reflectance products, which apply atmospheric correction for aerosols and water vapor *before* kelp classification. Images with more than 25% cloud cover are excluded, and remaining cloud and cloud-shadow pixels are masked using the quality-assurance band. Any localized change in atmospheric transparency, such as reduced sulfur aerosols from the fuel regulation, is therefore corrected at the reflectance stage rather than propagated into the kelp canopy estimates.

[3]Hexagonal grids are preferred over square or rectangular tessellations in spatial analysis because they provide equal area and equidistant centroids, minimize edge effects, and reduce sensitivity to the modifiable areal unit problem (Birch et al., 2007; Openshaw, 1984).

Vessel trajectories are derived from georeferenced ship positions recorded by the Automatic Identification System (AIS), made available by the U.S. Coast Guard.[4] We analyze the activity of cargo vessels (tankers, container ships, general cargo ships, and bulk carriers) at the hexagon–month level in terms of total distance traveled within the cell ("length", in meters) and count the number of unique vessel passages ("count").

More precisely, to assign shipping exposure to kelp locations, we use a standard Geographic Information System (GIS) method. For each hexagon containing kelp, we create a circular buffer of 20 km radius and identify all shipping hexagons that intersect this buffer. We then aggregate the shipping measures, summing length and count, across all identified shipping hexagons for each kelp hexagon i and quarter t. This yields two continuous exposure variables per kelp cell, which are the total distance traveled by vessels ($Length_{it}$) and the total number of vessel passages ($Count_{it}$).

$Length_{it}$ proxies the cumulative intensity of vessel activity, which may be linked, for instance, to discharges (exhaust emissions, antifouling leachates, greywater) that scale with distance traveled. In contrast, $Count_{it}$ captures the number of discrete vessel passages. This indicator is thus a complementary measure to the cumulative indicator by representing the frequency of "vessel-level events". These two computations may be interpreted as a way to distinguish between the intensive and the extensive margins of shipping exposure.

Environmental controls. As discussed in the Background section, kelp forests are affected by several environmental drivers, among which are sea surface temperature, wave intensity and precipitation (Carr and Reed, 2019; Castorani et al., 2022).

Sea surface temperature (SST) is sourced from MODIS monthly composites and aggregated to quarters. SST is a central determinant of kelp dynamics and is also informative about broader ocean conditions (e.g., Yang et al., 2023; Dunstan et al., 2018).[5]

Wave intensity is drawn from ERA5 (Hersbach et al., 2023) and measured as significant wave height (the average height of the highest third of waves), aggregated to quarters; it captures physical stress associated with storms and strong sea states.

Precipitation is drawn from ERA5 and aggregated to quarters; it proxies terrestrial nutrient runoff, which can affect kelp growth.

3.2 Specification

Our main design is a continuous difference-in-differences (see Callaway et al., 2024) based on the hypothesis that kelp cells that were more exposed to vessel traffic before 2009 should benefit more from the implementation of the regulated zone. The treatment intensity is measured by pre-policy shipping activity derived from AIS data as described in

[4] AIS data accessed via MarineCadastre.gov, jointly maintained by the U.S. BOEM and NOAA.

[5] We do not include a separate upwelling index (e.g., CUTI or BEUTI; Jacox et al., 2018) because SST is strongly negatively correlated with upwelling intensity along the California coast (cold surface water is the primary physical signature of upwelling events) and our year–quarter fixed effects already absorb the dominant seasonal upwelling cycle. Any remaining bias would require a differential change in upwelling at the policy date that is spatially correlated with shipping exposure.

the Data section, namely the log mean of vessel traffic (measured as total distance traveled) or the log mean of vessel count (over January–June 2009). Both are transformed as $\ln(1+x)$.

We estimate:

$$Kelp_{it} = \exp(\beta\, Exposure_i \times Post2009_t + \delta\, Z_{it} + f_i + f_t) \cdot \eta_{it}, \tag{1}$$

where η_{it} is a multiplicative error term satisfying $E[\eta_{it} \mid Exposure_i, Z_{it}, f_i, f_t] = 1$, and $Kelp_{it}$ is the sum of kelp biomass (kg) or canopy area (m^2) in hexagonal cell i during quarter t. $Exposure_i$ is the time-invariant, predetermined measure of shipping activity described above. $Post2009_t$ equals one from 2009Q3 onward and zero otherwise. Z_{it} is a vector of log-transformed environmental controls (log sea surface temperature, log precipitation, and log significant wave height).[6] f_i denotes kelp-cell fixed effects (one per cell) and f_t denotes year–quarter fixed effects. The cell fixed effects absorb any time-invariant determinant of kelp outcomes, including the main effect of $Exposure_i$, and the time fixed effects absorb shocks common to all cells, including the main effect of $Post2009_t$.

Because kelp outcomes are non-negative and include many zeros, we estimate Eq. (1) by Poisson pseudo-maximum likelihood (PPML; Santos Silva and Tenreyro, 2006).[7]

Standard errors are clustered at the cell level because adjacent cells may share ocean currents and temperature conditions. We also apply Conley spatial-HAC standard errors (Conley, 1999) at cutoffs of 100 km and 200 km, and find that the coefficient remains significant at the 1% level (Table 4 in Appendix B). The sample covers Southern California (latitudes 32°–36° N) from 2005 to 2016. The year–quarter fixed effects absorb any coast-wide ecological shock, including the 2013–2016 marine heat wave that caused catastrophic kelp losses in northern California (Rogers-Bennett and Catton, 2019; McPherson et al., 2021).

3.3 Results

Table 1 presents the results of our analysis, with the distance traveled as the exposure measure in Columns (1)–(2) and the vessel count in Columns (3)–(4). In each case, we report two kelp outcomes, the biomass and the canopy area.

[6] All controls are transformed as $\ln(1+x)$, which is a standard transformation in the field. For SST and wave height, which are strictly positive and large in magnitude, this is virtually identical to $\ln(x)$. For precipitation, the $\ln(1+x)$ transformation allows us to keep quarters with zero rainfall.

[7] We restrict the sample to the cells with vessel traffic within 20 km, defined as at least one vessel passage per month on average during January–June 2009. The excluded cells record less than one passage per month, and fewer than 0.35 for most of them. The estimates are unchanged when observations are weighted by log vessel count, and they remain positive and significant when observations are weighted by pre-period kelp biomass (Table 3, Panel D, in Appendix B).

	Distance Traveled		Vessel Count	
	(1) Biomass	(2) Area	(3) Biomass	(4) Area
Post 2009 × Exposure	0.1361***	0.1317***	0.1548***	0.1493***
	(0.0359)	(0.0352)	(0.0432)	(0.0424)
SST	-5.703***	-5.512***	-5.728***	-5.537***
	(1.145)	(1.136)	(1.150)	(1.142)
Precipitation	0.0764	0.0812	0.0846	0.0891
	(0.1003)	(0.0981)	(0.1012)	(0.0989)
Significant Wave Height	-3.267**	-3.102**	-3.358**	-3.189**
	(1.367)	(1.344)	(1.367)	(1.342)
Cell FE	Yes	Yes	Yes	Yes
Year-Quarter FE	Yes	Yes	Yes	Yes
Observations	3880	3880	3880	3880

Notes: Unweighted PPML estimates of Eq. (1). Kelp biomass is in kg and canopy area in m^2. Exposure is the log of mean distance traveled by vessels (Columns 1–2) or of mean vessel count (Columns 3–4) within 20 km of each kelp hexagon over January–June 2009. Environmental controls are in logs. Standard errors clustered by cell in parentheses.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 1: Continuous difference-in-differences: Kelp outcomes and the 2009 fuel rule

The main result of this analysis is that after the 2009 rule, kelp increased more in the cells that were more exposed to vessel traffic before the policy. The coefficient of interest (exposure interacted with the post-2009 dummy) is positive and significant whatever the specification.

In terms of impact, a doubling of pre-policy vessel traffic is associated with 10% more kelp biomass and canopy after 2009 when traffic is measured by distance traveled (Columns (1)–(2), $2^{0.136} - 1$), and about 11% when it is measured by vessel count (Columns (3)–(4)). Interestingly, as discussed in the Data section, the two measures describe different aspects of traffic (cumulative versus discrete impact), but they lead to the same conclusion.

Finally, environmental controls have the expected signs. Sea surface temperature has a negative effect, which is a well-known result (Carr and Reed, 2019; Rogers-Bennett and Catton, 2019), as is the negative impact of wave height, which causes physical damage to fronds and holdfasts during strong sea states.

The coefficient in Eq. (1) is based on a linear-in-log term that has been criticized recently. To avoid relying on that linear specification, we follow Callaway et al. (2024) and estimate the average *level* effect of high-exposure cells relative to low ones. We partition cells into exposure terciles, take the least-exposed tercile as the reference, and estimate group-specific post-2009 effects. We find that, after 2009, kelp in the most-exposed third of cells increased by roughly 46–51% relative to the least-exposed third (Table 5 in Appendix B). The effect is also positive for the middle third.

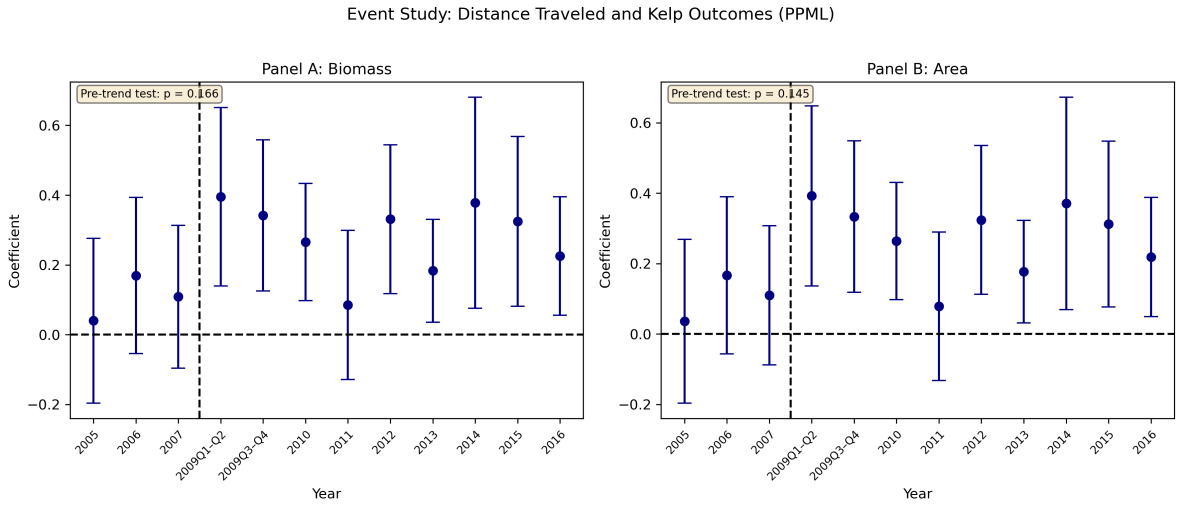
3.4 Dynamic effects

To assess whether the parallel-trends assumption holds and to trace the evolution of treatment effects over time, we estimate an event-study specification:

$$Kelp_{it} = \exp\left(\sum_{\ell \neq 2008} \beta_\ell \, Exposure_i \times \mathbf{1}_{t \in \ell} + \delta \, Z_{it} + f_i + f_t \right) \cdot \eta_{it}, \tag{2}$$

where ℓ indexes event-study periods and $\mathbf{1}_{t \in \ell}$ equals one when quarter t falls in period ℓ. Each calendar year constitutes a single period, except 2009, which is split into Q1–Q2 (pre-enforcement) and Q3–Q4 (post-enforcement). The reference period is 2008. This specification interacts the time-invariant treatment intensity with period indicators, allowing us to trace out the differential evolution of kelp outcomes across cells with different levels of shipping exposure.

Figure 1 reports the estimated coefficients $\hat{\beta}_\ell$ for distance traveled interacted with period indicators, using PPML with the same controls and fixed effects as previously.



Notes: The figure reports PPML estimates of the coefficients β_ℓ in Eq. (2), where exposure is the log of mean distance traveled by vessels within 20 km of each kelp hexagon over January–June 2009. The dependent variable is kelp biomass in Panel A and kelp canopy area in Panel B. Each period is a calendar year, except 2009, which is split into its first and second halves. The reference year is 2008. The vertical dashed line marks the beginning of 2009. Vertical bars show 95% confidence intervals based on standard errors clustered at the cell level. The box in each panel reports the p-value of a joint Wald test that the 2005–2007 coefficients are zero.

Figure 1: Event study: Distance traveled and kelp outcomes (PPML)

Two features of Figure 1 merit attention. First, all pre-policy coefficients (2005, 2006, and 2007) are close to zero and statistically insignificant, and a joint Wald test fails to reject the null of no pre-trend. Second, the coefficients turn positive and significant starting in 2009Q1–Q2, just six months before enforcement, and remain so through the end of the

sample (2016), with a temporary dip in 2011. The fact that the effect emerges already in 2009Q1–Q2 is consistent with some operators switching to compliant fuel during the compliance ramp-up period. We also run a placebo analysis by setting the treatment date at 2007Q3, and find no effect (Table 3, Panel E).

3.5 Threats to identification

Shipping exposure is (obviously) not randomly assigned across cells. Cells with heavy traffic differ from the others in several respects. For instance, as shown in Table 6 in Appendix C, they hold less kelp before the policy, and several confounding factors could vary with exposure over that period. We examine these potential problems in turn in what follows.

First, a stand that starts from a low level can grow faster in proportion than a dense one, for purely ecological reasons. This matters here because the specification is exponential, so β measures a proportional effect. Since high-exposure cells hold less kelp initially, such a recovery would produce a positive effect not related to the policy. Several additional analyses indicate that this possibility may not pollute our estimates too much (Table 3, Panel B). For instance, the correlation between exposure and baseline kelp is moderate (cell-level Pearson $r \approx -0.29$), so the two effects may be separated in the data. More importantly, the coefficient remains positive and significant when we control for the initial level of kelp. Finally, when the sample is split by baseline kelp, the effect is found in cells with *high* baseline kelp, where there is no depleted stand to recover.

A second potential bias may be due to the 2008–09 global financial crisis. Indeed, that shock reduced trade volumes in California ports by roughly 20% in 2009, the year the fuel rule came into force. A fall in vessel calls reduces traffic most in the cells that were most exposed, so part of the kelp gain could come from the fall in trade and not from the regulation. From our point of view, and for the ecological question, the origin of the shock does not really matter, since in both cases kelp responds to fewer vessels near the coast after 2009. But obviously, to evaluate the policy, we need to separate the two shocks. To do so, we construct a cell-level measure of port activity from the National Ballast Information Clearinghouse (NBIC) vessel arrival database and add it as a time-varying control. If the fall in trade explained the kelp gain, the exposure coefficient would decrease once port activity is held constant. It changes by at most 2% (Table 3, Panel C), which indicates that the estimated effect does not come from the variation in trade volumes.

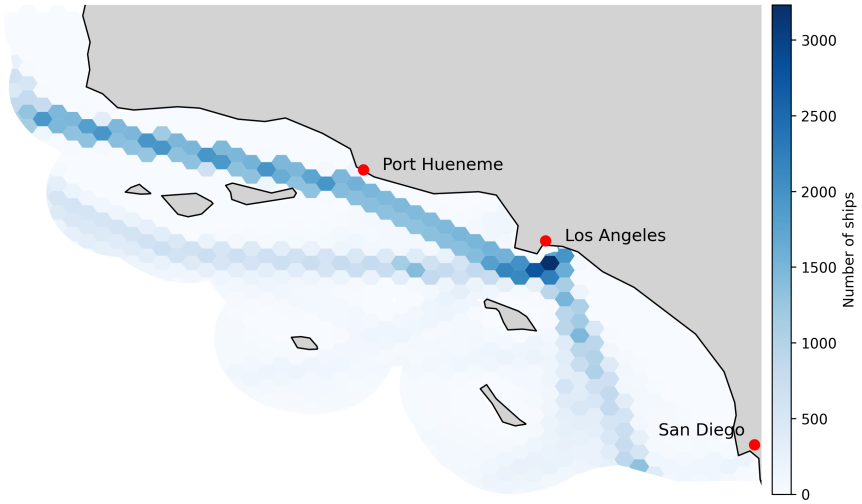
Finally, we should mention the boundary modification that occurred in December 2011 and that is analyzed in Section 4. Indeed, most of the post-2009 window falls after this modification, which changed traffic a second time in part of the same cells, so the post-2009 coefficient could combine the effects of the two policy changes. For this reason, we restrict the sample to 2005–2011, so that the entire post-period precedes the modification. All four coefficients remain significantly positive and have similar magnitudes to those in the full-sample analysis (Table 3, Panel C). This, at least, does not invalidate the previous analysis.

4 Boundary Modification of the Regulated Zone in 2011

We apply here the same empirical strategy as previously, but to the December 2011 boundary modification around the Channel Islands. That modification moved traffic back into specific corridors, so in that case we expect kelp to deteriorate in the cells most exposed to that traffic. That result of opposite sign after a policy change in the opposite direction is thus the second test of the relationship between maritime exposure and kelp.

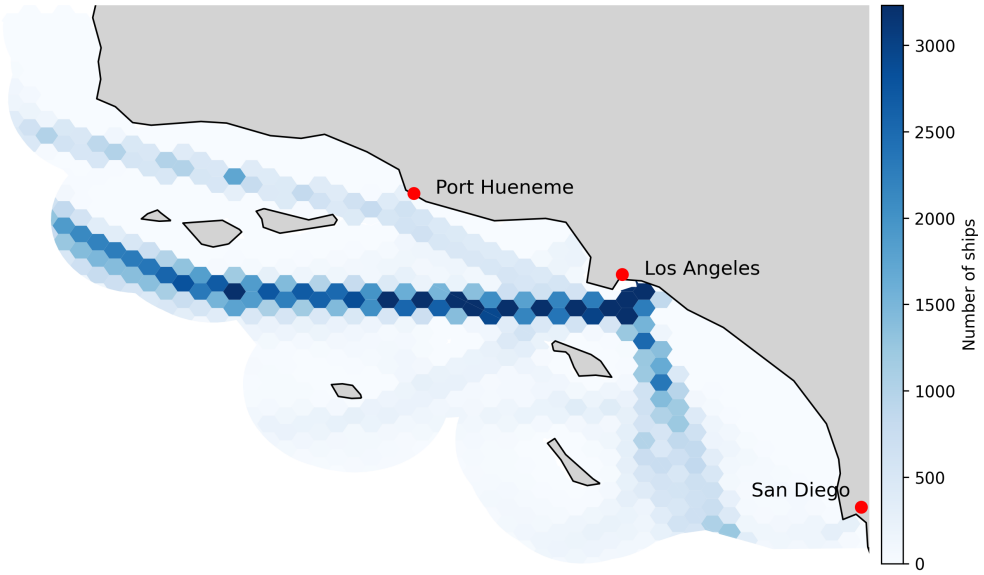
4.1 Spatial redistribution of traffic after 2011

Figures 2 and 3 map vessel traffic in 2009 and in 2011. In 2009, most traffic follows the lanes of the Santa Barbara Channel, along the mainland coast. In 2011, the main flow passes south of the northern Channel Islands, the route that vessels adopted to avoid the regulated zone (Klotz and Berazneva, 2022). The December 2011 modification extended the regulated zone to this southern route.



Notes: The map shows the number of vessel passages through each 10 km hexagonal cell of the Southern California coast in 2009. Darker shades of blue indicate more passages. Red dots mark the ports of Port Hueneme, Los Angeles and San Diego. Source: AIS data, authors' calculations.

Figure 2: Maritime traffic in Southern California, 2009



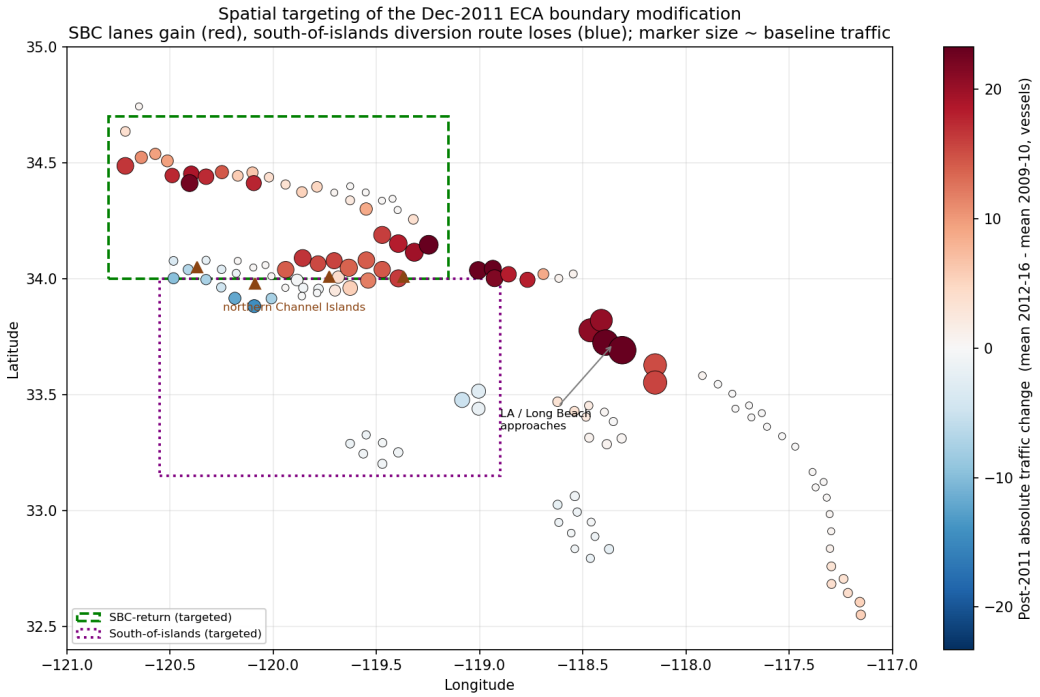
Notes: The map shows the number of vessel passages through each 10 km hexagonal cell of the Southern California coast in 2011. Darker shades of blue indicate more passages. Red dots mark the ports of Port Hueneme, Los Angeles and San Diego. Eleven of the twelve months shown precede the boundary modification of December 2011. Source: AIS data, authors' calculations.

Figure 3: Maritime traffic in Southern California, 2011

This extension should have brought traffic back near the kelp cells that were the most exposed in 2011. To check this, we regress the number of vessels within 20 km of each cell on the 2011 exposure interacted with a post-2011 dummy, with cell and year fixed effects.

We find that a doubling of exposure is associated with about 2.4 additional vessels after the modification.[8] Figure 4 shows where this change took place. Traffic increased in the Santa Barbara Channel lanes and decreased on the route south of the islands. A general recovery of shipping after the crisis would have raised traffic on both routes.

[8]The coefficient is 3.48 (standard error 0.56, clustered by cell; 889 cell-year observations, 2009–2016 excluding 2011).



Notes: The figure plots, for each kelp hexagon, the change in the mean number of vessels within 20 km between the pre-modification period (2009–2010) and the post-modification period (2012–2016). Red markers indicate an increase and blue markers a decrease. Marker size is proportional to mean traffic over 2009–2010. The dashed rectangle delimits the Santa Barbara Channel shipping lanes and the dotted rectangle the route south of the northern Channel Islands, which are shown as triangles. Source: AIS data, authors' calculations.

Figure 4: Spatial reallocation of vessel traffic after the 2011 boundary modification

4.2 Empirical strategy

We estimate the same continuous difference-in-differences as in Eq. (1), with exposure measured before the boundary modification. Kelp cells that were more exposed to vessel traffic before the modification should experience worse outcomes afterward if the return of traffic increased nearshore stressors.

We measure shipping exposure over the calendar year 2011, using total distance traveled and vessel count within a 20 km buffer around each kelp hexagon. The boundary modification took effect in December 2011, so eleven of the twelve months in the treatment variable reflect pre-modification routing.[9]

[9]Measuring exposure in 2010, a year that entirely precedes the modification, gives biomass coefficients of -0.077 (standard error 0.012) for distance and -0.213 (0.046) for vessel count, both significant at the 1% level.

We estimate:

$$Kelp_{it} = \exp\Big(\gamma\, Exposure_i^{2011} \times Post2011_t + \delta\, Z_{it} + f_i + f_t\Big) \cdot \eta_{it}, \tag{3}$$

where $Exposure_i^{2011}$ is the log of shipping intensity measured during 2011, $Post2011_t$ equals one from 2012Q1 onward and zero otherwise, and the other terms are defined as in Eq. (1). The equation is estimated by PPML with standard errors clustered at the cell level, on the same area and period as the 2009 specification (latitudes 32°–36° N, 2005–2016). A negative γ indicates that kelp deteriorated more after the modification in cells that were more exposed to shipping before it.

4.3 Results

Table 2 reports the results of this new analysis. We find that after the boundary change, kelp deteriorated more in the cells that were more exposed to shipping in 2011.

	Distance Traveled		Vessel Count	
	(1)	(2)	(3)	(4)
	Biomass	Area	Biomass	Area
Post 2011 × Exposure	-0.0925***	-0.0914***	-0.210***	-0.208***
	(0.0288)	(0.0282)	(0.0433)	(0.0425)
SST	-7.155***	-7.060***	-7.103***	-7.008***
	(0.904)	(0.909)	(0.903)	(0.908)
Precipitation	-0.0359	-0.0386	-0.0483	-0.0508
	(0.0781)	(0.0773)	(0.0782)	(0.0776)
Significant Wave Height	-3.452***	-3.371***	-3.341***	-3.262***
	(0.958)	(0.951)	(0.956)	(0.949)
Cell FE	Yes	Yes	Yes	Yes
Year-Quarter FE	Yes	Yes	Yes	Yes
Observations	6000	6000	6000	6000

Notes: Unweighted PPML estimates of Eq. (3). Kelp biomass is in kg and canopy area in m^2. Exposure is the log of mean distance traveled by vessels (Columns 1–2) or of mean vessel count (Columns 3–4) within 20 km of each kelp hexagon over 2011. Environmental controls are in logs. Standard errors clustered by cell in parentheses. Weighting observations by log vessel count gives biomass coefficients of -0.102 (standard error 0.025) for distance and -0.134 (0.047) for vessel count, both significant at the 1% level.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 2: Continuous difference-in-differences: Kelp outcomes and the 2011 boundary modification

The coefficient $\hat{\gamma}$ is negative and significant whatever the column considered. A doubling of pre-modification vessel traffic is associated with about 6% less kelp biomass and canopy after 2011 when traffic is measured by distance traveled, and about 13.5% less when it is measured by vessel count.

As expected, these results thus have the opposite sign to the 2009 estimates in Table 1, which confirms our initial analysis. The two policy changes moved maritime pressure in

opposite directions and kelp responded in opposite directions, which supports a causal interpretation of the relationship between vessel activity and kelp.

5 Conclusion

> "If in any country a forest was destroyed, I do not believe as many species of animals would perish as would here from the destruction of kelp." — Darwin (1839)

Human reliance on kelp dates back to prehistoric times; Erlandson et al. (2007) even propose a "Kelp Highway Hypothesis" explaining that human migration 20,000 years ago depended on this resource. Yet these same kelp corridors are now traversed by giant vessels that leave maritime pollution in their wake.

In this paper, we have exploited two policy episodes (the 2009 Ocean-Going Vessel Fuel Rule and the 2011 boundary modification around the Channel Islands) to estimate the causal effect of maritime regulation on kelp forests along the California coast, operating through differential shipping exposure. Using a continuous difference-in-differences design with shipping-specific exposure measures derived from AIS vessel trajectories, we show that (i) kelp canopy and biomass improved disproportionately in cells more exposed to vessel traffic after the 2009 regulation tightened fuel standards for nearshore vessels, and (ii) kelp deteriorated disproportionately in cells more exposed to shipping before the 2011 boundary change re-concentrated traffic in specific corridors. Event-study diagnostics show approximately flat pre-trends in the 2009 specification (Figure 1). The opposite signs of the 2009 and 2011 results (one following a tightening of fuel standards, the other following a spatial reallocation of traffic) strengthen the causal interpretation that maritime activity has a quantitatively important effect on kelp forest health, operating through channels that covary with the spatial footprint of maritime activity, which the regulation altered through both fuel switching and route adjustments.

These findings have implications for both economic analysis and maritime policy. First, as the international trade literature now examines the "hidden costs" of globalization, we consider that kelp forests deserve to be added to these computations due to the hidden impact of vessel traffic, and obviously because they play an important role in fisheries, alginate production, and carbon sequestration in coastal ecosystems.

Second, this study supports the objectives of the International Maritime Organization (IMO), which has been pivotal in enforcing ECAs over the past decade. In settings where regulated zones overlap with nearshore ecosystems, ECAs may have effects beyond their impact on air pollution, and could play a role in promoting sustainable international trade that limits damage to coastal habitats.

However, when implementing these policies, governments should keep in mind the spatial consequences of these regulations. Indeed, shipping is a derived demand of trade, so vessels always have to reach the ports whatever the regulation. An emission control area reduces the pollution of each ship, but its boundaries determine where the traffic goes. So the spatial design of the policy matters for the ecosystem of the underwater world.

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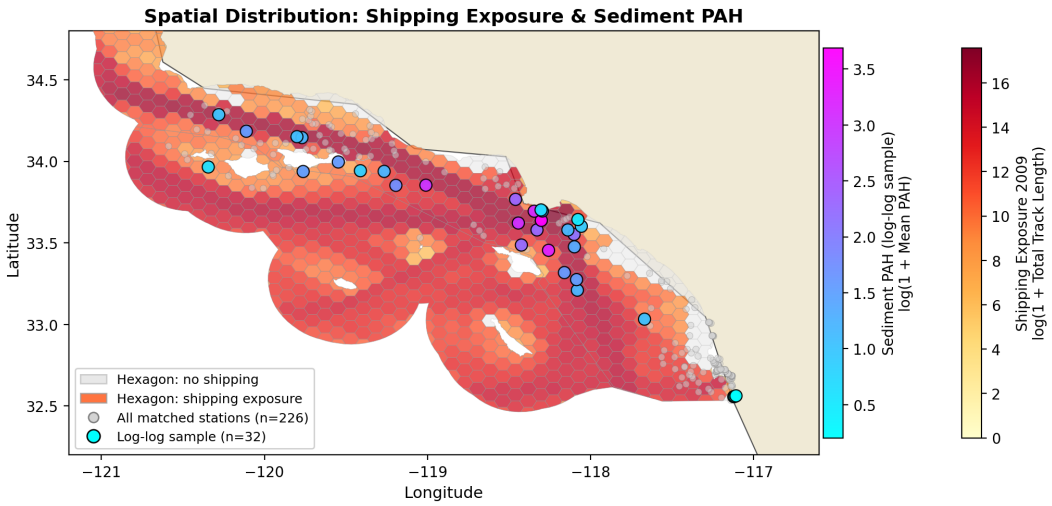
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A Pollution along shipping routes

Section 2 discusses several channels through which maritime activity may affect kelp. The effects of these channels scale with the concentration of vessel traffic. This appendix provides suggestive evidence that our AIS-based treatment variable captures variation in nearshore stressor exposure across kelp cells.

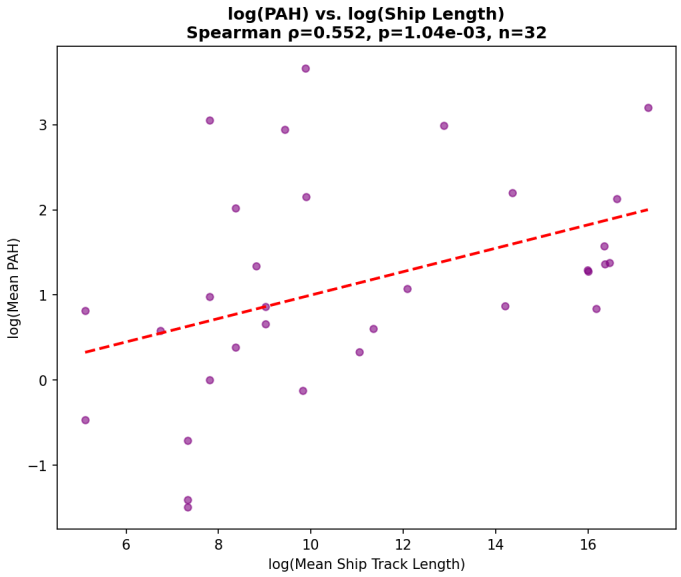
We match our AIS-derived shipping exposure to sediment chemistry stations from the Southern California Bight 2008 Regional Monitoring Program (Schiff et al., 2011), sampled before ECA enforcement. We focus on high molecular weight pyrogenic PAHs, which are produced by petroleum combustion (Yunker et al., 2002).



Notes: The 10 km hexagons show shipping exposure in 2009, measured as the log of one plus total vessel track length. Gray hexagons record no shipping. Dots are the 226 sediment stations of the Bight 2008 survey, each matched to the nearest hexagon within 10 km. The 32 colored dots are the stations used in Figure 6, colored by the log of one plus mean PAH concentration. Source: AIS data and Bight 2008 survey, authors' calculations.

Figure 5: Shipping exposure and sediment PAH concentrations in the Southern California Bight

Stations with the highest pyrogenic PAH concentrations are found in areas of high pre-policy vessel traffic (Figure 5). Across the 32 Bight stations with positive PAH concentrations and shipping exposure, log pyrogenic PAH concentrations are positively correlated with log mean vessel track length (Pearson $r = 0.41$; Spearman $\rho = 0.55$; Figure 6).



Notes: Each point is one of the 32 stations of Figure 5. The vertical axis gives the log of mean PAH concentration in the sediment (Bight 2008 survey) and the horizontal axis the log of mean vessel track length in the matched hexagon (2009). The dashed line is the OLS fit, and the title reports Spearman's rank correlation with its p-value. Source: AIS data and Bight 2008 survey, authors' calculations.

Figure 6: Log-log relationship between shipping exposure and sediment PAH concentrations

B Robustness of the 2009 Estimates

Table 3 reports the coefficient on shipping exposure interacted with the post-2009 indicator under alternative specifications of Eq. (1). Panel A reproduces the main estimates of Table 1. Panels B and C correspond to the three factors examined in Section 3.5.

In Panel B, we add controls built from the pre-policy level of kelp to account for the recovery of initially depleted stands: (i) log pre-policy biomass interacted with the post indicator, (ii) baseline-kelp quartiles interacted with the post indicator, (iii) log pre-policy biomass interacted with a linear trend, and (iv) log pre-policy biomass interacted with a full set of year effects. We also estimate the main specification separately on cells above and below the median of pre-policy biomass.

Panel C contains two specifications. The first adds port activity as a control for the 2008–09 financial crisis. Port activity is measured as the log of the inverse-distance-weighted sum of quarterly overseas vessel arrivals at Los Angeles, Long Beach, San Diego and Port Hueneme, from the National Ballast Information Clearinghouse.[10] The second restricts the sample to 2005–2011, the years before the 2011 boundary modification.

In Panel D, observations are weighted by log vessel count or by pre-period kelp biomass (2005–2008 cell average). Panel E assigns a fictitious treatment date of 2007Q3 on the pre-policy sample (2005–2008).

[10] Data available at `https://nbic.si.edu/`.

	Distance Traveled		Vessel Count		
	Biomass	Area	Biomass	Area	Obs.
Panel A: Main specification					
Baseline	0.136***	0.132***	0.155***	0.149***	3880
	(0.036)	(0.035)	(0.043)	(0.042)	
Panel B: Recovery of depleted kelp stands					
+ ln(baseline kelp) × post	0.103***	0.099***	0.114***	0.109***	3880
	(0.038)	(0.037)	(0.043)	(0.042)	
+ baseline-kelp quartile × post	0.122***	0.118***	0.136***	0.130***	3880
	(0.040)	(0.039)	(0.047)	(0.046)	
+ ln(baseline kelp) × trend	0.121***	0.117***	0.135***	0.130***	3880
	(0.036)	(0.035)	(0.042)	(0.041)	
+ ln(baseline kelp) × year effects	0.084**	0.080**	0.095**	0.090**	3880
	(0.037)	(0.036)	(0.042)	(0.041)	
High-baseline cells only	0.129***	0.124***	0.145***	0.139***	1920
	(0.038)	(0.038)	(0.046)	(0.045)	
Low-baseline cells only	0.014	0.012	0.026	0.023	1960
	(0.104)	(0.104)	(0.112)	(0.112)	
Panel C: Financial crisis and 2011 boundary modification					
+ port activity (NBIC arrivals)	0.136***	0.132***	0.152***	0.146***	3880
	(0.036)	(0.035)	(0.043)	(0.043)	
Pre-2011 window (2005–2011)	0.123***	0.122***	0.137***	0.135***	2268
	(0.041)	(0.041)	(0.050)	(0.050)	
Panel D: Weighting					
Weighted by log vessel count	0.135***	0.131***	0.152***	0.146***	3880
	(0.037)	(0.037)	(0.045)	(0.044)	
Weighted by pre-period kelp biomass	0.103***	0.100***	0.124***	0.120***	3736
	(0.030)	(0.029)	(0.038)	(0.036)	
Panel E: Placebo					
Fake treatment date 2007Q3 (2005–2008)	-0.075	-0.069	-0.086	-0.080	1248
	(0.067)	(0.067)	(0.075)	(0.075)	

Notes: Each entry is the PPML coefficient on exposure interacted with the post-2009 indicator in Eq. (1), estimated with the change given in the row label. All regressions include the environmental controls in logs and kelp-cell and year–quarter fixed effects. Baseline kelp is the mean biomass of the cell over 2005–2008. The row weighted by pre-period kelp biomass and the placebo row exclude the 3 cells with no kelp over 2005–2008. Standard errors clustered by cell in parentheses.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: Robustness of the 2009 estimates

When standard errors allow for correlation between cells up to 100 km or 200 km apart, the exposure coefficient remains significant at the 1% level in the four columns (Table 4).

	Distance Traveled		Vessel Count	
	(1) Biomass	(2) Area	(3) Biomass	(4) Area
Post 2009 × Exposure	0.1361	0.1317	0.1548	0.1493
Cell-clustered SE	(0.0359)***	(0.0352)***	(0.0432)***	(0.0424)***
Conley SE (100 km)	[0.0405]***	[0.0398]***	[0.0478]***	[0.0473]***
Conley SE (200 km)	{0.0296}***	{0.0293}***	{0.0370}***	{0.0369}***
Cell FE	Yes	Yes	Yes	Yes
Year-Quarter FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Observations	3880	3880	3880	3880

Notes: The point estimates are those of Table 1. The rows below report the cell-clustered standard error in parentheses and the Conley spatial-HAC standard errors with cutoffs of 100 km in brackets and 200 km in braces, computed with a Bartlett kernel and great-circle distances. Observations of the same cell are at zero distance from each other, so the Conley standard errors allow for any correlation within a cell over time. Stars give the significance level implied by each standard error.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Spatial-HAC (Conley) standard errors

In Table 5, the linear exposure term is replaced by indicators for exposure terciles and quartiles, with the least-exposed group as the reference. All the other groups gain kelp after 2009 relative to this reference, and the gain is significant for the most-exposed group in all columns.

	Distance Traveled		Vessel Count	
	Biomass	Area	Biomass	Area
Panel A: Exposure terciles (base = lowest tercile)				
Middle tercile	0.598***	0.585***	0.275*	0.266
	(0.105)	(0.104)	(0.165)	(0.162)
Top tercile	0.413***	0.400***	0.390***	0.379***
	(0.151)	(0.150)	(0.146)	(0.144)
Panel B: Exposure quartiles (base = lowest quartile)				
Second quartile	0.109	0.110	0.057	0.058
	(0.186)	(0.185)	(0.167)	(0.166)
Third quartile	0.541***	0.522***	0.526***	0.507***
	(0.146)	(0.144)	(0.146)	(0.144)
Top quartile	0.379***	0.367***	0.365**	0.352**
	(0.143)	(0.142)	(0.143)	(0.142)
Linear coefficient	0.136	0.132	0.155	0.149

Notes: Each entry is the PPML coefficient on an exposure-group indicator interacted with the post-2009 indicator. Cells are grouped by terciles (Panel A) or quartiles (Panel B) of 2009 exposure, and the least-exposed group is the reference. All regressions include the environmental controls in logs and kelp-cell and year–quarter fixed effects. The last row reports the coefficient of Table 1 for comparison. Standard errors clustered by cell in parentheses.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Nonparametric dose-response: group-specific post-2009 effects

C Balance Table

Table 6 reports pre-treatment (2005–2008) means of key variables across quartiles of the treatment variable (log vessel distance traveled, January–June 2009). The last column reports the p-value from a Kruskal–Wallis test for differences across quartiles. Mean sea surface temperature and precipitation are close across quartiles, and significant wave height is lowest in the most-exposed quartile. All three are controlled for in the regression. Pre-treatment kelp biomass and canopy area are highest in the least-exposed quartile; Appendix B reports specifications that control for baseline kelp.

	Q1 (low)	Q2	Q3	Q4 (high)	KW p
Log vessel distance (Jan-Jun 2009)	9.311	10.491	12.470	13.321	0.000
	[0.491]	[0.290]	[0.564]	[0.182]	
Kelp biomass (kg)	1655116.8	801416.6	409706.0	321770.5	0.000
	[3976548.5]	[2459511.5]	[929510.4]	[904773.3]	
Kelp canopy area (m^2)	220694.5	107794.3	54520.6	43562.9	0.000
	[521297.4]	[333245.2]	[121129.7]	[123756.5]	
Log SST	2.842	2.883	2.798	2.844	0.000
	[0.134]	[0.124]	[0.125]	[0.124]	
Log precipitation	2.611	2.506	2.779	2.729	0.009
	[1.052]	[1.116]	[1.221]	[1.164]	
Log significant wave height	0.8859	0.7991	0.8732	0.7286	0.000
	[0.1626]	[0.1313]	[0.1748]	[0.1465]	
Log vessel count (Jan-Jun 2009)	0.9722	1.516	3.593	4.300	0.000
	[0.1562]	[0.310]	[0.468]	[0.134]	
Cells	21	20	20	20	

Notes: Cell-quarter means over 2005–2008, by quartile of exposure, measured as the log of mean distance traveled by vessels within 20 km of each kelp hexagon over January–June 2009. Standard deviations in brackets. The last column reports the p-value of a Kruskal–Wallis test of equal distributions across quartiles. Sample: the cells of Table 1.

Table 6: Pre-treatment balance across treatment-intensity quartiles